kept clean and maintained in excellent condition. Equipment that has dirty or malfunctioning parts shall not be used until the parts are cleaned, repaired or replaced as necessary, to the satisfaction of the Quality Manager and Ministry Representative.

Paint shall be applied in a uniform layer, with overlapping at the edge of the spray pattern. The spray pattern shall be adjusted so that the paint is deposited uniformly. During application, the gun shall be held perpendicular to the surface and at a distance that will ensure that a wet layer of paint is deposited on the surface. The trigger of the gun shall be released at the end of each stroke.

All runs and sags shall be brushed out immediately, or the paint shall be removed and the surface repainted.

Areas inaccessible by spray such as cracks and crevices and the blind sides of rivets and bolts shall be painted by brush. Stripe coat material may be applied by spray, to speed up the transfer of material to the surface, but all stripe coating must be brushed in to ensure complete coverage.

Fluid tips shall be of proper orifice size and fan angle, and fluid control gun of proper construction as recommended by the manufacturers of both the material being sprayed and the equipment being used.

The air pressure to the paint pump shall be adjusted so that the paint pressure to the gun is proper for optimum spraying effectiveness. The pump pressure shall be controlled by a suitable regulator with working gauges.

Pressures higher than that necessary to properly atomize the paint shall not be used.

The air compressor used shall be of sufficient size to maintain the regulated air pressure to the pump constant at all times irrespective of the amount of equipment being operated from the compressor.

Proper filters shall be provided in the high pressure line so that dirt, dry paint or other foreign materials are not deposited on the paint film.

The trigger of the gun shall be pulled fully open and held fully open during all spraying to ensure proper application of paint. The trigger shall be released at the end of each stroke.

216.10.5 Mitt and Dauber Application - Mitt and Dauber application shall be limited to situations where other coating application methods are poorly suited. Their use shall be as specified by the Designer and only with prior acceptance by the Ministry Representative.

Coating thickness shall be uniform over the entire surface of the object coated. Coating shall be free of pinholes or other holidays, sags, etc.

216.10.6 Powder Coatings - Powder coatings are finely ground coatings that are electrostatically sprayed onto the substrate, then melted into a consolidated layer by heat. Coatings may be thermoplastic or thermosetting. Cleaning is typically in a series of chemical baths.

Substrates contaminated with soluble salts shall be washed to remove those salts before being cleaned to meet specification. Acceptable levels of soluble salts shall be at the NV 2 level as stated in SSPC-SP 12.

The substrate to be coated shall be cleaned to the coating manufacturer's specifications. The water test shall show a clean surface.

If required for adhesion, a conversion coating shall be applied to the substrate before powder coating.

Fresh and reclaimed powder shall be dry, free flowing, and free of lumps and contaminants.

Soaking oven temperatures and residence times shall be sufficient to completely melt the powder, but not enough to oxidize or discolour the coating.

Coating thickness and gloss shall be uniform over the entire surface of the object coated. Coating shall be free of pinholes or other holidays.

216.11 Application Methods (Metallizing and Hot Dip Galvanizing)

216.11.1 Metallizing - Metallizing shall be applied in accordance with CSA-G189

Metallizing includes the thermal spraying of zinc or Zn:Al/85:15 alloy onto a properly prepared metal substrate.

The method of heating the spray metal is either by flame or by plasma arc. Spray metal may be in wire or powder form to fit the apparatus used to spray the metal. The application method shall ensure that a uniform, clean, well adhered, film of metal is applied to the steel surface.

Surface preparation shall be SSPC SP5 white metal blast. Profile shall be sharp, 50 to 75 μm.

Metallizing shall be applied by qualified personnel with documented field experience metallizing steel structures such as bridges.

Metallizing shall be applied in a uniform layer at a thickness of 150 to 200 µm thickness. For special applications, the thickness may be specified outside this range.

When the metallizing is to be overcoated to form a duplex system, the newly metallized surface shall be protected from contamination until coated. If the metallized surface becomes contaminated, the steel substrate shall be re-cleaned to SSPC SP5/NACE No. 1 as above and re-metallized.

Metallizing is porous, and application of a coating over the surface may require a mist coat or other special procedures to avoid bubbling and subsequent problems caused by the interaction of the porous coat with the coating being applied.

Metallizing may be applied over a much wider range of ambient conditions than normal paint. When application conditions may deviate outside normal coating conditions, the Contractor shall consult the Quality Manager and Ministry Representative for acceptance before proceeding.

216.11.2 Hot Dip Galvanizing - Hot dip galvanizing shall be applied per CAN/ CSA-G164.

216.12 Application of Coatings

216.12.1 General - Application of coatings on steelwork specified to be painted as part of the Contract or the Design shall meet the following requirements.

216.12.1.1 Welding Requirements - Areas to be field welded shall have no paint applied for a distance of at least 100 mm back from the weld area.

If inorganic zinc primer is used, the set-back shall be a minimum 25 mm.

All welds shall be cleaned of weld residue including weld spatter and roughness on the welds. Sharp edges shall be rounded to a minimum radius of 1 mm.

All welds shall receive a stripe coat of midcoat material over the primer prior to the midcoat.

Welding done on existing coated surfaces shall have the coating removed, past the heat affected zone or 100 mm from the weld whichever is more, down to bare metal. Surface profile of the cleaned area shall be at least 25 µm (1mil).

216.12.1.2 Faying Surfaces - Unless specified otherwise in the Design, the faying surfaces of weathering steel shall be blast cleaned but not coated.

When faying surfaces are specified to be coated, faying or friction connections shall be coated with only a prime coat of a Class B certified coating. A copy of the Class B test certificate shall be supplied to the Quality Manager and the Ministry Representative for each coating used on a faying surface. The class of coating is as defined in CSA/CAN S6-00.

Coating and curing conditions shall strictly follow all application parameters listed on the Class B Certificate. Failure to do so will require re-cleaning and recoating according to the Class B parameters for that coating.

216.12.1.3. Surfaces Inaccessible After Erection - Surfaces inaccessible after erection shall receive the field coats of paint in the shop, before shipment to the erection site. Prior to erection, handling damage shall be repaired to the satisfaction of the Quality Manager and Ministry Representative.

216.12.1.4 Application and Curing of Successive Coats of Paint - Each coat of paint shall be in a proper state of cure or dryness before the application of the succeeding coat. Unless otherwise specified by the Designer a minimum of overnight drying shall be required. Under poor drying and curing conditions, this period will be extended.

Curing conditions shall be those given by the Coating Manufacturer in the product application instructions. Curing time will vary according to temperature and in some cases humidity. The required curing time is estimated at the ambient conditions during curing. This will normally be done from data supplied by the Coating Manufacturer.

Coatings that have a maximum overcoating window shall not be overcoated after that time without appropriate surface preparation as provided in writing by the Coating Manufacturer. The overcoat window will be calculated from data supplied by the Coating Manufacturer using the ambient conditions at the site.

216.12.1.5 Primer - Primer shall not be applied until the cleaned steel surface meets specification requirements. Once specification requirements are met for a substrate, the primer must be applied within 8 hours. The substrate if protected will be assumed to continue to meet the specification for 8 hours unless obvious contamination has occurred. This will be at the discretion of the Quality Manager and the Ministry Representative.

Primer shall be applied to specification requirements. Should the specification requirements be different than the Coating Manufacturer's application instructions then the more stringent specification shall apply unless otherwise approved by the Ministry Representative.

216.12.1.6 Penetrating Sealers - Penetrating sealers or low viscosity products designed to wick into seams and seal them shall be applied according to the Coating Manufacturer's product data sheets unless the Design calls for specific application parameters. Should that be the case, the sealer shall be applied to the Design requirements.

Current industry thinking has these products applied on top of a single primer coat and under stripe and midcoats.

Under certain circumstances it may be necessary to use a penetrating sealer as a primer or as a first coat. This will be specified by the Designer and subject to acceptance by the Ministry Representative.

216.12.1.7 Stripe Coat - All areas such as crevices, corners, rivets, bolts, nuts, welds, edges and other protrusions shall be given a stripe coat to ensure that there is sufficient coating dry film thickness. Coating may be applied by spray, but it shall be brushed into all areas to be striped.

The stripe coat is a separate coat and shall be applied at the thicknesses given for the midcoat coating.

The stripe coat shall be applied to the area and at least 25 mm out from the area to be striped.

The stripe coat shall be a contrasting colour to the coats above and below it.

The stripe coat shall be cured before being overcoated.

216.12.1.8 Midcoat - Midcoat shall be applied as a full coat onto clean, cured, primer, sealer, and stripe coats. The stripe coat and or sealer shall be cured before being coated.

216.12.1.9 Caulking - Caulking shall be applied after the midcoat and prior to the topcoat.

Caulking shall be compatible with the coating system used.

Silicone Caulking shall not be used.

Suitable caulking materials are listed in DBSS 308 under the sub heading System PCM Paint Caulking Materials.

Caulking shall be applied as specified in the Contract.

Caulking shall not be topcoated until it cures according to the Manufacturer's product data sheets. Lower temperatures may increase this time.

If there is a time window for top coating, the caulking shall be topcoated before the window expires.

216.12.1.10 Topcoat - Once the undercoat and caulking if any have cured, the topcoat shall be applied.

216.12.1.11 Lapping Existing Coatings - Primer if applied shall not overlap existing coatings by more than 10-20 mm. Ministry preference is that it just contacts the existing feathered coating or overlaps very slightly to seal the join.

Midcoat shall overlap existing coatings by a minimum 50 mm.

Topcoat shall overlap existing coatings by a minimum 50 mm and shall hide all newly added midcoat material.

216.12.2 Shop Coating

216.12.2.1 General - Unless otherwise specified the following procedure shall apply to all new steel prior to shipment to the site of erection:

Before blast cleaning, the steel shall be checked for contamination of non-visible salts and the surface shall meet SSPC-SP12/NACE No. 5, NV-2.

Before coating, the steel shall be blast cleaned to SSPC-SP 10/NACE No.2 Near white. Abrasive blast cleaning will be permitted prior to fabrication provided that any contamination incurred during fabrication is removed prior to coating, leaving an SSPC-SP 10/NACE No.2

surface. Small areas of contamination occurring during fabrication shall be cleaned by manual cleaning to SSPC-SP 3 Power tool cleaning and/or SSPC SP-12/NACE No. 5 LPWC as necessary to remove contamination. Large areas of contamination occurring during fabrication shall be dry abrasive blast cleaned again to SSPC-SP 10/NACE No.2 Near-white blast.

Surface profile for primers shall be 50 to 75 μm and sharp.

Prior to dry abrasive blasting, all welding slag, spatter and alkaline deposits in the vicinity of the welds shall be removed. Surfaces shall then be blast cleaned to SSPC-SP 10/NACE No.2 Near-white blast.

When fabrication is completed and the surfaces cleaned as required, they shall be inspected by the Quality Manager and Ministry Representative prior to the application of paint.

After passing inspection and before any contamination has occurred, prime all steel surfaces with one coat of the selected shop coat primer from DBSS 308. Primer shall be applied to specification requirements. Should the specification requirements be different than the Coating Manufacturer's application instructions then the more stringent specification shall apply unless otherwise approved by the Ministry Representative.

A dry abrasive blast cleaned surface shall be coated within 8 hours of cleaning and before rust blooming becomes apparent. If either condition is exceeded then the area must be re-blasted to specification.

Clean or partially coated steel shall be protected from contamination including, but not limited to dirt, oil, grease, soluble salts and overspray, while in the shop. Contaminated surfaces shall be re-cleaned by the Contractor.

Surfaces that will be in contact with concrete shall be coated with the full system prior to erection. Coatings with known incompatibilities with high pH shall not be used in contact with the concrete.

Surfaces inaccessible after erection shall receive the field coats of paint in the shop, before shipment to the erection site.

If more than one coat is to be shop applied, manufacturer's recommended recoat times for the ambient temperatures shall be followed.

Generally, a minimum of 16 hours drying/curing time shall be required between the application of any paint in the shop and the moving or handling of any piece so painted. This period may be increased in cases where poor drying conditions exist, such as shop areas open to the weather in winter months. Since curing times vary with different coating products, variations from this time may occur with specific products.

Quick Dry/Cure products or products with "winter" catalysts shall not to be substituted for DBSS 308 products unless specified by the Designer and accepted by the Ministry Representative.

216.12.3 Transport of Coated Items - Transport of coated items shall be done in a manner that imparts minimum damage to the coating system. The coatings shall be cured before handling and shipment.

Canvas or Fabric slings shall be used to move the items. Chains or other methods that are known to damage coatings shall not be used.

Coated items to be transported shall have sufficient dunnage to prevent coating damage. Transport tie-downs shall be padded to protect the coated surfaces and edges.

Transport damage shall be repaired by the Contractor.

216.12.4 Field Repair of Damaged Shop Coating and Field Weld Areas - After the complete erection of shop coated steel members the following procedure shall apply:

Before coating, field weld areas shall be ground off to remove all welding slag, spatter and alkaline deposits in the vicinity of the welds. Washing will be required when the shop coat or steel has been contaminated with air borne dirt, dust, salt, chemicals, etc., unless the steel shows no visible contamination, and meets the SSPC SP-12/NACE No. 5, NV2 levels for non-visible contaminants. Surfaces shall then be cleaned to SSPC-SP 11 Power Tool Cleaning to Bare Metal. The existing coating shall be feathered back 25 mm from damaged areas into good coating. Feathering shall continue until the existing coating is well adhered. This may be considerably more than the 25 mm feather back.

Care shall be taken not to damage surrounding areas.

Note: This cleaning shall include the 100 mm welding set back.

All areas of the shop applied coating system which have been damaged during handling, shipping and erection and all field connections shall be cleaned to SSPC-SP 3 Power tool clean or SSPC SP11 Power tool cleaning to bare metal at the discretion of the Quality Manager and Ministry Representative. Low pressure water cleaning shall be used as necessary to remove dirt, salts and other deleterious contamination.

All areas that have been cleaned to bare metal as required shall be coated with an organic zinc field primer, applied by brush to ensure full coverage. If the area is too large to coat practically with a brush, then spray may be used, providing overspray does not contaminate the surrounding finished coating.

Lapping of new and existing coatings shall be as DBSS 216.12.1.11 above.

If a partial system was applied in the shop then the primed areas shall be coated with successive coats to the level of the surrounding coating then the entire structure shall be topcoated. This treatment shall include a brushed-in stripe coat.

If a coating system was completely applied in the shop and on erection some damage has occurred or there are still uncoated areas, then these areas shall be cleaned per this section above and successively coated with each coat of the system up to the topcoat including a brushed-in stripe coat. Manufacturer's recoat times for the ambient temperatures shall be followed.

216.12.5 Field Topcoat - Remaining coats shall be applied as per DBSS 216.13 below.

216.12.5 Field Coating

216.13.1 Touch-up coating This section deals with the repair of small areas of coating by the Contractor. The definition of small areas, will vary depending on the situation, but in general, will apply to areas that can be economically treated using the touch-up technique.

The coating used for touch up and repair shall be one of the DBSS 308 Field Overcoat (FO) systems.

The area to be repaired and coated shall be washed to SSPC SP-12/NACE No. 5, WJ-4, with a surface cleanliness of NV-2 before attempting further cleaning. Areas of the coating that have failed or rusted shall be cleaned to SSPC-SP 3 or SSPC-SP 11 using the appropriate tools.

The old coating shall be feathered back 25 mm from damaged areas into good coating. Feathering shall continue until the coating is well adhered. This may be considerably more than the 25 mm feather back.

Pack rusted crevices will require more thorough cleaning. The degree of cleaning required will be detailed in the Design for each specific structure. After cleaning, they shall be coated with the prime coat, and then a brush coat of penetrating sealer shall be applied to the crevice and allowed to soak in. Repeated applications of the penetrating primer may be necessary until the inside of the crevice is saturated. Where the coating system used does not include a specific primer coat, the penetrating sealer shall be applied as a first coat.

In general, coating used for touch up and repair shall be of the same type as those used initially. For example if the coating system to be touched up is an alkyd system, then alkyds are the coatings of choice. The coating to be used shall be accepted by the Ministry Representative and as specified in the Design.

The selected primer shall be applied in the damaged areas. The primer shall be cured for the time recommended by the manufacturer prior to additional coating being applied

After the primer is cured, the midcoat(s) shall be applied to bring the surface of the coating to the same level as the surrounding sound coating. The midcoat shall be applied within the manufacturer's suggested range of thickness. The midcoat(s) shall be cured for the time(s) recommended by the manufacturer.

After the midcoat material is cured, the area shall be given a topcoat. Where practical, the topcoat shall be extended to a logical edge such as the edge of the beam or plate, to present a less patchy appearance.

Lapping of new and existing coatings shall be as DBSS 216.12.1.11.

216.13.2 Overcoating - The object is to clean the surface, repair spot or area damage with coatings and to overcoat the entire area with a topcoat.

The coating used for touch up and repair shall be one of the DBSS 308 Field Overcoat (FO) systems.

The area to be repaired and coated shall be washed to SSPC SP-12/NACE No. 5, WJ-4, with a surface cleanliness of NV-2 before attempting further cleaning. Areas of the coating that have failed or rusted shall be cleaned to SSPC-SP-3 or SSPC-SP-11. Larger failed areas shall be cleaned to SSPC-SP 6/NACE No.3 Commercial blast clean. In this case, surface profile should be 50-75 µm and sharp.

Debris contaminating the rest of the structure or the immediate environment due to cleaning operations shall be removed by the Contractor.

The old coating shall be feathered back 25 mm from damaged areas into tightly adhered coating. If the coating cannot be removed with a dull putty knife, it is considered tightly adhered.

Pack rusted crevices shall be cleaned as per DBSS 216.13.1 above.

Once all the existing damaged areas are built up to the level of the existing coating, the topcoat is applied over the entire area to seal the existing coating and all the repairs.

In general, coating used for overcoating shall be of the same type as those used initially. For example if the coating system to be touched up is an alkyd system, then alkyds are the coatings of choice. Use of an over wintered test patch is recommended to test compatibility. The coating to be used shall be accepted by the Ministry Representative and as specified in the Design.

216.13.3 Recoating - The area to be recoated shall be washed to SSPC SP-12/NACE No. 5, WJ-4, with a surface cleanliness of NV-2 before attempting further cleaning. The entire area shall be blast cleaned to SSPCSP 6 Commercial blast clean. Note that some specialty coatings or coatings in some difficult applications such as immersion may require SSPC-SP 10/NACE No. 2 Near-white blast or even SSPC-SP 5/NACE No. 1 White metal blast.

Surface profile should be 50 - 75 µm and sharp.

The Contractor shall provide full containment of debris generated by the cleaning, coating and curing operations.

Debris contaminating the rest of the structure or the immediate environment due to cleaning operations shall be removed by the Contractor.

Sharp edges shall be radiused to 1 mm.

216.13.4 Zone Coating - Zone coating may use any one of or a combination of touch-up, overcoating or recoating on limited areas of a structure.

216.13.5 Field Curing of Coatings - Each coating application shall be cured before it is coated over or exposed to weather or traffic.

Coatings inside enclosed spaces shall continue to be ventilated until dry or cured to ensure that they cure correctly.

Uncured coating shall be protected against damage from dust or other detrimental foreign matter. If necessary, coating shall be delayed until the cause of the dust, etc. has abated. Uncured coating shall be protected from the weather until cured. In some cases, this may include hoarding, heating, ventilation or dehumidification. Coating contaminated or damaged by any of the above conditions shall be repaired by the Contractor.

216.14 Damage to Adjacent Property - It is the intention of the Ministry of Transportation that the coating work shall have no adverse effect on adjacent property. Repair of damage to adjacent property, vehicles, pedestrians and other portions of the structure due to the Contractor's coating operations is the sole responsibility of the Contractor.

216.15 Cleanup - All construction and painting debris must be removed from the site and the site environs. The site and surroundings shall be left in the same condition as prior to work being commenced.

216.16 Not Used

SECTION 218

CONCRETE FOR MINOR WORKS

Note: For any SS Drawing (SP XXX - XX) referenced in this Section refer to the like-numbered Drawings in the "Standard Specifications For Highway Construction" 2006 edition.

218.01 General - DBSS 218 shall apply only when DBSS 218 has been specifically specified for a cast in place concrete component by the Ministry in the Contract. Otherwise cast in place concrete work shall comply with DBSS 211.

The Work shall be carried out in accordance with the best modern practice of cement concrete construction.

The Designer, Quality Manager and Ministry Representative shall determine the suitability of the materials, their quantity, quality and condition; of the Contractor's plant, equipment and personnel; of the time and manner of carrying out the work and of every operation and matters incidental thereto.

218.02 Cement - Portland cement shall conform to CSA Standard A5 - Portland Cement

218.03 Aggregates - The aggregates shall be the best procurable and shall comprise inert, inorganic mineral matter only. They shall be satisfactorily clean, durable, hard, sound, strong and tough and be practically free from alkali, clay, dust, loam, mica, shale, silt or any material deemed objectionable in nature or amount by the Quality Manager and Ministry Representative. None of the particles comprising the aggregates shall be coated nor be unreasonably flat or elongated. No aggregate may be used while frozen.

218.04 Grading of Aggregates - The grading of all aggregates shall be such as will produce an economic, plastic, practical, smooth, workable mix which when placed, set and cured, will produce a concrete of the best character possible.

To secure such grading, the Quality Manager or Ministry Representative may require the Contractor to modify the aggregate by the addition or subtraction of proportions of specific sizes of material.

218.05 Classes of Concrete - The classes of concrete are A, B, C, X and Y. Each class of concrete shall be used only when it is called for in the Contract. Unless otherwise specified, Class "A" Concrete shall be used.

Concrete deposited in water shall be Class "A" with 10% additional cement.

218.06 Classification and Proportioning of Concrete Mixtures - The materials shall be proportioned so that the amount of cement contained in 1 m^3 of concrete in place shall be approximately as shown in Table 218-A.

The proportions of water, cement, coarse and fine aggregate to be used shall be determined by the Designer . Measurement shall be made by mass, and the accuracy must be satisfactory to the Quality Manager; all such allowances for contained water, bulking or otherwise must be made as found necessary to allow for the proportions ordered by the

TABLE 218-A CLASSIFICATION AND PROPORTIONING OF CONCRETE MIXTURES

CLASS OF CONCRETE	REQUIRED STRENGTH AT 28 DAYS MPa	SIZE OF COARSE AGGREGATE (mm)	CEMENT (40 kg) SACKS PER CUBIC METRE	MAXIMUM WATER CONTENT PER SACK OF CEMENT (ℓ)
A	20	25.0 to 5.0	9.1	21.0
B	15	50.0 to 5.0	7.0	24.5
C	10	63.0 to 5.0	5.6	30.0
X	17.5	50.0 to 5.0	9.2	24.0
Y	20	20.0 to 5.0	9.8	21.0

Designer, being accurately provided in each batch.

218.07 Machine Mixing - The mixing of concrete Quality Manager shall be done in a batch mixer of type approved by the Quality Manager. The mixer shall be equipped with a suitable charging hopper, water storage and water measuring device. It shall be cleaned at frequent intervals when in use and maintained in such a condition that the mixing action will be unimpaired.

218.08 Duration of Mixing - The mixing of each batch shall continue not less than 90 seconds after all the materials are in the mixer, during which time the mixer shall rotate at a speed of from 14 to 20 revolutions per minute unless otherwise rated by the manufacturer.

218.09 Hand Mixing - When hand mixing is authorized by the Designer or Quality Manager, it shall be done on a watertight platform. The fine aggregate shall be spread on the platform and the cement spread evenly over the fine aggregate. The material shall then be shovelled into a cone shaped pile by casting centrally on the pile. This pile shall then be divided by casting into two or more cone shaped piles and the operation of dividing and reuniting continued until the batch is uniform in colour. Only sufficient water to produce the desired consistency shall then be added by sprinkling as the batch is turned. The coarse aggregate previously moistened shall then be mixed with the mortar in the manner specified for mixing fine aggregate and cement.

218.10 Retempering - The retempering of concrete or mortar which has partially hardened, that is, remixing with or without additional cement, aggregate or water, will not be permitted.

218.11 Forms - Forms shall be so constructed that the finished concrete will conform to the shapes, lines, grades and dimensions indicated on the Design.

Lumber used in forms shall be free from warp. For exposed surfaces, it shall be dressed to a uniform width and thickness and be free from loose knots, decay or other defects. For unexposed surfaces and rough work, undressed lumber may be used if means are taken to prevent leakage of mortar.

Lumber once used in forms shall have nails withdrawn and surfaces to be in contact with concrete thoroughly cleaned and smoothed before being used again.

Unless otherwise specified, suitable moulding or bevels shall be placed at the angles of forms to round or bevel the edges and re-entrant angles of the concrete.

For internal ties, bolts and rods shall preferably be used. They shall be so arranged that when the forms are removed no metal shall be within 25 mm of any exposed surface. Wire ties may be permitted only on light work; they shall not be used through surfaces where discolouration will be objectionable.

The inside of forms shall be coated with non-staining mineral oil, or other approved liquid or thoroughly wetted, (except in freezing weather). Where oil is used it shall be applied before the reinforcement is placed.

Care shall be taken to ensure that forms do not become dried and warped before concrete is deposited. Before concrete is placed, forms and reinforcement shall be checked and approved by the Quality Manager

Forms shall not be disturbed until the concrete has adequately hardened. The proper time for removal of forms shall be determined by the Designer.

Care shall be taken when removing forms to avoid roughening or injuring corners and to keep edges sharp.

Whenever forms are removed from showing faces before the concrete has become hard and dry, the surface of the concrete shall be immediately wetted and kept wet for at least three days. Defects in showing faces shall be rectified to the satisfaction of the Quality Manager and Ministry Representative.

218.12 Depositing Concrete - Concrete shall be deposited in forms in approximately uniform horizontal layers as near as practicable in its final position to avoid re-handling. Depositing shall be continuous until the unit of operation is complete and as rapid as practicable to ensure bonding of the successive layers.

Concrete shall be thoroughly worked around reinforcement and embedded fixtures and into all parts of the forms. Coarser particles shall be worked back from the face of the forms.

218.13 Not Used

SECTION 232

METAL BIN-WALL

Note: For any SS Drawing (SP XXX - XX) referenced in this Section refer to the like-numbered Drawings in the "Standard Specifications For Highway Construction" 2006 edition.

232.01 Scope - This Section covers the construction of metal bin type retaining walls in accordance with the general layout and details indicated on SS Drawings 1-SP232 through 3-SP232.

232.02 Provision of Metal Bin-Wall - Bin-wall of the design (depth to height ratio) called for shall be constructed at the locations and as shown on the Design with the materials, accessories, and necessary ancillary work all in accordance with the details indicated on the Design or as directed by the Designer.

232.03 Materials - Bin-wall material will be supplied by the Contractor in accordance with the Design.

Structure backfill shall be well-graded granular material with a maximum size of 75 mm and not more than 8% passing a 0.075 mm (#200) sieve.

232.04 Construction

232.04.01 Construction - Construction shall be carried out with all labour, tools, equipment and incidentals necessary to complete all bin-walling in accordance with good work practice.

232.04.02 Foundation - Foundation excavation shall be carried down to the depths shown on the Design or to such greater depths as the Designer may direct. In the case where rock, hardpan or other unyielding material is encountered, it shall be removed to a depth 300 mm below the design grade and backfilled in accordance with DBSS 232.04.04.

232.04.03 Erection - The units shall be assembled as shown on the Design. Components shall be handled carefully and any which are damaged shall be removed and replaced.

Before the assembly bolts are tightened, columns shall be checked for line and batter or plumbness, where applicable.

232.04.04 Backfill - Structure backfill material shall be placed as wall foundation, backfill inside the bins, and for 0.6 m behind the wall. The material shall be spread in layers not exceeding 150 mm in thickness and shall be compacted to a minimum 95% of the laboratory density obtained by the current ASTM test method D 698.

Backfill inside each bin shall progress simultaneously but not in advance of that behind the wall.

Note: Bin wall may be backfilled as it is erected provided that the columns are held in correct position while the backfill is being placed.

232.05 Not Used

SECTION 303

CULVERTS

Note: For any SS Drawing (SP XXX - XX) referenced in this Section refer to the like-numbered Drawings in the "Standard Specifications For Highway Construction" 2006 edition.

DESCRIPTION

303.01 Scope - This Section covers the construction of pipe culverts where shown on drawings or as directed by the Designer, and in accordance with the Contract, the drawings and Special Provisions covering the various types. The trench and other preparatory work shall be approved by the Quality Manager before actual placing starts.

Culvert materials are normally durable under most conditions. Factors which affect the service life of culvert material are corrosion from acidic and alkaline conditions in the soil, abrasion from bedload carried by the stream and freeze/thaw cycles. Plastic culverts shall not be exposed to direct sunlight (ultra violet rays) or installed in areas where there is a potential fire hazard.

Filtration control devices, such as filter cloth, must not be attached to the end of culverts.

Obstructed culverts shall be cleared as outlined in DBSS 165.10.07.

All culvert pipe, unless otherwise specified, will be supplied by the Contractor to the requirements of the Contract, generally in accordance with DBSS 145.15.

MATERIALS

303.11 Concrete Pipe - Reinforced concrete pipe supplied will have joints of the tongue and groove type all in accordance with the current specification ASTM C 76 Series, and plain concrete pipe in accordance with the current specification ASTM C 14 series.

303.12 Corrugated Steel Pipe - Corrugated steel pipe supplied will conform to the requirements of DBSS 320.

303.13 Polyethylene Pipe – The materials for pipe and fitting shall be high-density polyethylene meeting ASTM D 3350 Cell Classification 335420C.

303.14 PVC Pipe – See DBSS 318.03

303.15 Structural Plate Corrugated Steel Pipe/Pipe-Arch (Field Assembled) Culverts - Special care shall be taken in the installation of all structural plate corrugated steel culverts.

CONSTRUCTION

303.31 Concrete, Corrugated Steel and Plastic Pipes – This Subsection does not cover Structural Plate Corrugated Steel Pipe/Pipe Arch installations.

303.31.01 Trenching - A trench shall be excavated to the depth and grade as given by the design. A full trench condition shall be provided wherever possible, but in no case shall the trench be of lesser depth than 50% of the diameter of the pipe. If, in the opinion of the Quality Manager, the material in the bottom of the excavation is of such a character as to cause unequal settlement along the length of the culvert, the trench shall be dug below the grade to such depth as ordered, backfilled with gravel or other suitable material, and thoroughly tamped or otherwise compacted to ensure a firm and uniform foundation.

Where rock, in either ledge or boulder formation, occurs in the bottom of the trench, the rock shall be excavated below grade and backfilled with sand or fine gravel so that there will be at least a 150 mm cushion between the rock and all portions of the pipe.

The bottom of the trench upon which the pipe is to be laid shall be shaped so that at least 25% of the circumference of the pipe is in contact with the prepared foundation for the whole of its length.

303.31.02 Placing - Concrete pipe shall be laid beginning at the lower end with the groove and pointing up-grade. Pipe with elliptical reinforcement shall be laid with the minor axis of the reinforcement as marked by the manufacturer in a vertical position.

When jointing concrete pipes the trench shall be in a dry condition and the joints shall be cleaned and wetted before sealing with mortar. The mortar shall consist of one part Portland Cement to two parts fine sand, mixed to the proper consistency. Sealing shall be neatly and thoroughly done, and the interior of the pipe cleaned of all surplus mortar. Joints shall be kept damp with burlap or earth for at least three days after sealing.

Corrugated steel pipe shall be laid beginning at the lower end with the outside laps pointing up-grade and

lower end with the outside laps pointing up-grade and the longitudinal joint on the side. The separate section shall be firmly jointed together and any steel in joints, which is not thoroughly protected by galvanizing, shall be coated with a suitable asphaltum paint.

303.31.03 Backfilling - Backfill and Bedding Material shall consist of mineral aggregate and shall meet the gradation specified in Table 303-A.

For metal structures with corrugation depths greater than 63 mm, the backfill maximum particle size may be as high as one-half the corrugation depth or 75 mm whichever is smaller.

Minimum bedding thickness is 150 mm or twice the corrugation depth whichever is greater.

For embankment installation, bedding and backfill will extend two-thirds of the span or 0.9 m beyond the culvert span on either side, whichever is greater. For trench installation, bedding and backfill will extend a minimum of 0.3 m on either side. Backfill will extend to the base of the pavement structure (to the surface in the case of high fines gravel surface) or to a depth of cover of two-thirds of the span (diameter) whichever is less. Bedding and backfill shall be placed in layers not exceeding 150 mm in depth. Backfill shall be placed symmetrically. The differential backfill height on either side of the culvert shall not exceed 0.3 m. Bedding and backfill shall be compacted to a minimum 95% of the laboratory density by following the current ASTM D 698, except the 0.3 m layer immediately above the pipe which shall be placed with no compaction.

TABLE 303-A GRADATION REQUIREMENTS FOR CULVERT BACKFILL AND BEDDING MATERIAL

SIEVE SIZE mm	PERCENTAGE PASSING
31.5	100
25	60-100
19	15-100
2.36	10-100
0.075	0-5

303.31.04 End Walls - End walls shall be built around the ends of pipe culverts as shown on the Design or as directed by the Designer. The construction of these end walls shall meet the requirements governing the construction of Class "A" concrete (See DBSS 218.05).

303.31.05 Inlet and Outlet Ditches - Inlet and outlet ditches to culverts shall be constructed to the lines and levels as given by the design, field-fit as necessary. The excavation shall be classified as "Roadway and Drainage Excavation," not "Foundation Excavation."

303.32 Structural Plate Corrugated Steel Pipe/Pipe-Arch Culverts

303.32.01 Foundation Excavation - If a coarse boulder material solid rock, or other unsuitable foundation is encountered when preparing the bed of the pipe, the material shall be excavated a minimum depth of 300 mm below the pipe and backfilled with granular material having a maximum size of 75 mm.

303.32.02 Backfilling - All backfill materials for a minimum width of 3 m on either side of the pipe and to a depth of 1 m above the pipe, if grade allows, shall be granular material as specified by the design and approved by the Quality Manager.

End dumped or loose pushed material shall not be piled closer than 3 m from the pipe. Spreading in layers not exceeding 150 mm and compaction shall be carried out by equipment operating parallel to the pipe. The area close to the pipe shall be compacted by small mechanical tampers. No boulders in the backfill material of a size greater than 75 mm shall be closer than 1.2 m to the pipe surface.

Extreme care must be taken to ensure the backfilling operation is carried out equally on both sides of the pipe. Hauling equipment shall not be operated over the pipe until backfill operations have completed a minimum cover of 600 mm over the pipe.

SECTION 308

COATING SYSTEMS

308.01 Scope - This Section defines the coating systems required for shop and field coating of structural steel or wood/concrete surfaces which may be exposed to exterior corrosive conditions.

308.03 Standards and Specifications

a)Specifying Bodies: SSPC The Society for Protective Coatings**,** has a large number of coating and cleaning specifications. **ASTM** The American Society for Testing and Materials, produced test procedures for a wide range of products and processes. **NACE,** The National Association of Corrosion Engineers, also produces specifications and test procedures for coatings and corrosion engineering. All have a web presence available by searching the name above.

b)Colour Specifications using a five digit number refer to the US Federal Specification 595b colour numbers.

c)Measuring Units

The metric unit (μm) or imperial unit (mil) can be used interchangeably, at the conversion factor of:

1 mil = 25 μm (1 mil = 1 thousandth of an inch)

308.04 System Selection Table - This table summarizes the coating systems currently in use. The table lists application areas where specified coating systems may be used. It is intended to assist in the choice of a coating system for specific applications. Some systems may find use in more than one specific application or in applications not mentioned.

Where a system is specified in the Contract or on the Design, only those approved coating systems listed for that system may be used.
The generic coating systems below mirror the same categories in the Recognized Products List, which lists approved coatings in each of the categories below.

COATING SYSTEM SELECTION TABLE (PART 1)

System Code	Application Area	Structure Environment	Coating System Primer - Midcoat - Topcoat
	STRUCTURAL STEEL		
SS1	Steel, Shop	Standard New Steel Coating for Moderate to Severe Service	Inorganic zinc, epoxy, polyurethane
SSWP	Steel, Shop Weldable Primer	Under Development	
SF1	Steel, Field	Severe Field Application	Metallizing, epoxy, polyurethane
SF2	Steel, Field	Standard Field Recoat for Moderate to Severe Service	Organic zinc, epoxy, polyurethane
SF3	Steel, Field	Severe Where Adjacent buildings/traffic etc require a Dry Fall Material	Metallizing, vinyl, vinyl
SF4	Steel, Field	Moderate to Severe Where Adjacent buildings/traffic etc require a Dry Fall Material	Organic zinc, vinyl, vinyl
SF5	Steel, Field	Moderate	Alkyd on steel
SF6	Steel Field	Moderate to Severe Field Application	Multi-Component Urethane Systems
SFIM1	Steel Field, Immersion	Moderate to Severe Field Application	Coal Tar Epoxy or Equivalent

System Code	Application Area	Structure Environment	Coating System Primer - Midcoat - Topcoat
SFO1	Steel, Field Overcoat	Moderate conditions	Water-based Acrylics and Vinyls
SFO2	Steel, Field Overcoat	Under Development	Water Based Acrylic Epoxy
SFO3	Steel, Field Overcoat	Moderate conditions	Solvent Based Single Component
SFO4	Steel, Field Overcoat	Moderate to Severe	Solvent Based Surface Tolerant Epoxy
SFO5	Steel, Field Overcoat	Moderate	Alkyd on steel CAN/CGSB-1.59M alkyds
SFAG	Steel, Field Antigraffiti Coatings	Under Development	Varies with system

COATING SYSTEM SELECTION TABLE (PART 2)

System Code	Application Area	Structure Environment	Coating System Primer - Midcoat - Topcoat
SFTB	Steel, Field Touchup Brushable	Small Damaged Areas, or as a Holding Primer	Rust Inhibitive Primer / Topcoat
SFTA	Steel, Field Touchup Aerosol	Small Damaged Areas, or as a Holding Primer	Rust Inhibitive Primer / Topcoat
	UNIQUE SITUATIONS		
U1	Steel, Unique Products	Moderate	Rust Inhibitive Sealer / Topcoat
U2	Steel, Unique Products	Low Temperature, high humidity	Organic Zinc / Midcoat / Topcoat
	PENETRATING SEALERS		
SPS	Steel, Penetrating Sealers	Penetrating Sealers for Joints, Bolts, Rivets etc.	Various Unique Systems, Designed To Be Used With Coating Materials From The Same Manufacturer
	CAULKING MATERIALS		
PCM	Paint Caulking Materials	For Non-immersion Service	Assorted
	WEATHERING STEEL		
SW *	Steel, Weathering Field Coatings	Under Development	Two or three coat systems
	CONCRETE		
CF1	Concrete, Field	Under Development	Cementatious
CF2	Concrete, Field	Under Development	Non-Cementatious
CFAG	Concrete, Field Antigraffiti	Under Development	Concrete Antigraffiti
	HOT DIPPED GALVANIZING		
GO	Hot Dip Galvanizing, Overcoat	Under Development	
	STEEL RAILINGS		

SRG	Steel, Hot Dip Galvanizing	Preferred Treatment for Removable Railings	Hot Dip Galvanizing
SRM	Steel, Metallizing	Primarily for non-removable posts	Zinc - Zinc Aluminum Powder or Wire
	WOOD		
WFR	Wood, Railings and Rubrails	Moderate to Severe	Primer, Midcoat, Topcoat CAN/CGSB-1.59M alkyds
WF1	Wood, Fencing	Under Development	Oil based sealer 1-GP-145M
WF2	Wood, Fencing	Under Development	Latex Based Sealer 1-GP-204M
WFT	Wood, Field Touchup	Moderate to Severe	Primer, Midcoat, Topcoat CAN/CGSB-1.59M alkyds
	INLAND FERRIES		
InFer	Steel Inland Ferries	Under Development	

308.05 Not Used

308.06 System Selection Criteria - The selection of a coating system is dictated by the location and environment of the structure to be coated. The coating system may be applied in a shop or in the field. The field location may necessitate a specific maintenance regimen, for example, a complete recoat or a single overcoat. The environment may be moderately or severely corrosive. In addition, there may be concerns with limiting overspray.

The following general comments apply to all coatings supplied.

- Coatings shall not be mixed and matched between suppliers, or within a supplier's product line.

- All coatings shall bear the manufacturer's name and address, batch number, paint type, date of manufacture, DOH or US Federal Specification 595b color code, and mixing and application instructions. All labels shall contain the requisite WHMIS label information and be accompanied by a valid MSDS. All coatings are subject to inspection and testing by the Quality Manager or Ministry Representative before acceptance or at the time of application.

308.07 Coating Systems - The Ministry Recognized Products List (RPL) lists coating products that have been reviewed and accepted for use on Ministry projects. The RPL can be reviewed at the following Ministry web site:

http://www.th.gov.bc.ca/publications/eng_publications/geotech/Recognized_Products_Book.pdf

Products not on the Ministry Recognized Products List shall not be used without the prior acceptance of the Designer and the Ministry Representative. Unless noted otherwise in this Section, products on the RPL shall be used in accordance with the introduction/definitions, descriptions, notes, qualifiers and all other information given in the RPL. The use of the products from the RPL by the Contractor shall in no way relieve the Contractor from its contractual obligations.

The following pages list requirements for systems appearing in DBSS 308.04 - Coating System Selection Table.

STEEL, SHOP - SYSTEM SS1: Inorganic Zinc primer, Epoxy midcoat, Polyurethane topcoat

MoT Usage: This system is designed for use in a shop coating environment where the new steel is protected from the weather during coating and curing. The system will provide good service in a moderate to severe field environment.

PRIMER	SEALCOAT	STRIPECOAT	MIDCOAT	2nd MIDCOAT	TOPCOAT
Inorganic Zinc	None	Epoxy Midcoat Material	Epoxy Midcoat Material	None	Aliphatic Polyurethane
SSPC-SP 10/NACE No. 2, "Near-White Metal", Sharp Profile: 50 - 75 µm (2 - 3 mils)					

Notes for use:

- Dry film thicknesses shall be as per the manufacturer's application instructions unless otherwise approved by the Designer and the Ministry Representative.
- Depending on the system chosen, the inorganic zinc primer may require a mist coat to prevent bubbling in the mid-coat material.
- Inorganic zinc shall be dry and cured before additional coatings are applied.
- Stripe coating shall be applied on top of the primer and brushed to a minimum stripe width of 50 mm (2 inches).
- While most of the above systems require SSPC-SP 10 surface preparation, SSPC-SP 5 "White Metal" blasting will typically give a longer service life.

STEEL, FIELD - SYSTEM SF1: Metallizing primer, Epoxy midcoat, Polyurethane topcoat

MoT Usage: This system is designed for use in a field coating environment where extreme exposure such as splash zones on through trusses, fence posts (panels should be hot dip galvanized), bearings, expansion joints and any steel exposed to de-icing chemical solutions dripping from above; structures adjacent to pulp mills or smelters are present.

PRIMER	SEALCOAT	STRIPECOAT	MIDCOAT	2nd MIDCOAT	TOPCOAT
Zinc wire or powder 150-200 µm (6 - 8 mils)	15% thinned epoxy Midcoat Material	None	Epoxy Midcoat Material	None	Aliphatic Polyurethane
SSPC-SP5/NACE No. 1, "White Metal", Sharp Profile: 50 - 75 µm (2 - 3 mils)					

Notes for use:

- Dry film thicknesses shall be as per the manufacturer's application instructions unless otherwise approved by the Designer and the Ministry Representative. The metallizing is normally 6 to 8 mils, but may in extremely severe environments be made thicker to enhance the lifetime.
- The sealcoat is applied to partially fill the voids in the metallizing and reduce bubbling in the subsequent coat.
- Previously galvanized railings, fence posts, and panels shall be "Brush-off" cleaned and top-coated with a different colour than the structure, after a wash primer is applied.

- Metallizing can be either zinc or zinc:aluminum (85:15) alloy wire thermal spray, flame process, plasma arc process or spray and fuse powder process.
- If field steel has been exposed to weather, deicing salts or dirt, an SSPC- SP 12/NACE No. 5, LP WC water wash shall be done before blasting to remove salts to an SSPC- SP 12/NACE No. 5, NV-2 level.

STEEL, FIELD - SYSTEM SF2: Organic Zinc primer, Epoxy midcoat, Polyurethane topcoat

MoT Usage: This system is designed for use in a field coating environment that is moderate to severe**. This is the most often used field coating system.**

PRIMER	SEALCOAT	STRIPECOAT	MIDCOAT	2nd MIDCOAT	TOPCOAT
Organic Zinc Primer	May use a Penetrating Sealer	Epoxy Midcoat Material	Epoxy Midcoat Material	None	Aliphatic Polyurethane
SSPC-SP6/NACE NO. 3, "Commercial Blast", Sharp Profile: 50 - 75 µm (2 - 3 mils)					

Notes for use:

- Dry film thicknesses shall be as per the manufacturer's application instructions unless otherwise approved by the Designer and the Ministry Representative.
- If field steel has been exposed to weather, deicing salts or dirt, an SSPC- SP 12/NACE No. 5, LP WC water wash shall be done before blasting to remove salts to an SSPC- SP 12/NACE No. 5, NV-2 level..
- Stripe coating shall be applied on top of the primer and brushed to a minimum stripe width of 50 mm (2 inches).

STEEL, FIELD - SYSTEM SF3: Metallizing primer, Vinyl midcoat, Vinyl topcoat

MoT Usage: This system is designed for use in a field coating environment where extreme exposure areas such as splash zones on through trusses, fence posts (panels should be galvanized), bearings, expansion joints and any steel exposed to de-icing chemical solutions dripping from above and no adjacent pulp mills or smelters are present. This system shall be used only when epoxies cannot be used due to adjacent buildings, traffic, etc. This system is high VOC and should only be used where other coating systems such as SF1 cannot be used.

PRIMER	SEALCOAT	STRIPECOAT	MIDCOAT	2nd MIDCOAT	TOPCOAT
Zinc wire or powder 15 - 200 µm (6 - 8 mils)	May use a thin coat of Vinyl Midcoat Material	None	Vinyl Midcoat Material	None	Vinyl Topcoat Material
SSPC-SP 5 "White Metal", Sharp Profile: 50 - 75 µm (2 - 3 mils)					

Notes for use:

- Dry film thicknesses shall be as per the manufacturer's application instructions unless otherwise approved by the Designer and the Ministry Representative. The metallizing is normally 6 to 8 mils, but may in extremely severe environments be made thicker to enhance the lifetime.
- The sealcoat is applied to partially fill the voids in the metallizing and reduce bubbling in the subsequent coat.
- Previously galvanized railings, fence posts, and panels shall be "Brush-off" cleaned and top-coated with a different colour than the structure, after a wash primer is applied.
- Metallizing can be either zinc or zinc:aluminum (85:15) alloy wire thermal spray, flame process, plasma arc process or spray and fuse powder process.
- If field steel has been exposed to weather, deicing salts or dirt, an SSPC- SP 12/NACE No. 5, LP WC water wash shall be done before blasting to remove salts to an SSPC- SP 12/NACE No. 5, NV-2 level.

STEEL, FIELD - SYSTEM SF4: Organic Zinc primer, Vinyl midcoat, Vinyl topcoat

MoT Usage: This system is designed for use in a field coating environment that is moderate to severe. This system shall be used only when epoxies cannot be used due to adjacent buildings, traffic, etc. (e.g., pedestrian overpasses).

This system is high VOC and should only be used where other coating systems such as SF2 cannot be used.

PRIMER	SEALCOAT	STRIPECOAT	MIDCOAT	2nd MIDCOAT	TOPCOAT
Organic Zinc Primer	None	Vinyl Midcoat Material	Vinyl Midcoat Material	None	Vinyl Topcoat Material
SSPC-SP6/NACE NO. 3, "Commercial Blast", Sharp Profile: 50 - 75 µm (2 - 3 mils)					

Notes for use:

- Stripe coating shall be applied on top of the primer and brushed to a minimum stripe width of 50 mm (2 inches).
- If field steel has been exposed to weather, deicing salts or dirt, an SSPC- SP 12/NACE No. 5, LP WC water wash shall be done before blasting to remove salts to an SSPC- SP 12/NACE No. 5, NV-2 level.

STEEL, FIELD - SYSTEM SF5: Alkyd System on Steel

MoT Usage: This system is designed for use in a field coating situation where more moderate conditions are expected. It may also be used in areas where surface preparation is less than ideal.

PRIMER	SEALCOAT	STRIPECOAT	MIDCOAT	2nd MIDCOAT	TOPCOAT
Alkyd Primer CAN/CGSB 1.40-89	None	Alkyd Midcoat Material	Alkyd Midcoat CAN/CGSB 1.59-89M	None	Alkyd Topcoat CAN/CGSB 1.59-89M
SSPC-SP 2 "Hand Tool Cleaning" or SSPC-SP 3 "Power Tool Cleaning"					

Notes for use:

- While this system will work on the cleaning regimen listed above it will perform better on SSPC- SP 6 “Commercial blast”.
- Stripe coating shall be applied on top of the primer and brushed to a minimum stripe width of 50 mm (2 inches).
- If field steel has been exposed to weather, deicing salts or dirt, an SSPC- SP 12/NACE No. 5, LP WC water wash shall be done before blasting to remove salts to an SSPC- SP 12/NACE No. 5, NV-2 level.

STEEL, FIELD - SYSTEM SF6: Multi-Component Urethane Systems

MoT Usage: This system is designed for use in a field coating environment in moderate to severe exposure areas.

PRIMER	SEALCOAT	STRIPECOAT	MIDCOAT	2nd MIDCOAT	TOPCOAT
Zinc Primer Material		Midcoat Material	Midcoat Material		Topcoat Material
High Pressure Water Wash then SSPC- SP 2 "Hand Tool Cleaning" or SP 3 "Power Tool Cleaning"					

Notes for use:

- While this system will work on the cleaning regimen listed above it will perform better on SSPC- SP 6 Commercial blast.
- Stripe coating shall be applied on top of the primer and brushed to a minimum stripe width of 50 mm (2 inches).
- If field steel has been exposed to weather, deicing salts or dirt, an SSPC- SP 12/NACE No. 5, LP WC water wash shall be done before blasting to remove salts to an SSPC- SP 12/NACE No. 5, NV-2 level.

STEEL, FIELD IMMERSION - SYSTEM SFIM1: Coal Tar Epoxy or Equivalent

MoT Usage: This system is designed for use in a field coating environment in immersion situations

PRIMER	SEALCOAT	STRIPECOAT	MIDCOAT	2nd MIDCOAT	TOPCOAT
None Typically		Topcoat Material Per Manufacturers Instructions	Topcoat Material Per Manufacturers Instructions		Topcoat Material Per Manufacturers Instructions
Water Wash as below then per the Manufacturer's Instructions Typically SSPC-SP 5 blast					

Total Thickness: Per Manufacturer's Instructions or Specification

Notes for use:

- Some coal tar filled products are carcinogenic. Read the MSDS for detailed information
- If field steel has been exposed to weather, deicing salts or dirt, an SSPC- SP 12/NACE No. 5, LP WC water wash shall be done before blasting to remove salts to an SSPC- SP 12/NACE No. 5, NV-2 level.

STEEL, FIELD OVERCOAT - SYSTEM SFO1: Water-based Acrylic or Vinyl System

MoT Usage: This system is designed for use in a field over coating situation where the original alkyd system is intact or where loose rust/paint can be removed by hand/power tools before repair and topcoating.

PRIMER	SEALCOAT	STRIPECOAT	MIDCOAT	2nd MIDCOAT	TOPCOAT
Primer Material	As Per Manufacturer's Instructions	As Per Manufacturer's Instructions	As Per Manufacturer's Instructions		As Per Manufacturer's Instructions
High Pressure Water Wash then SSPC- SP 2 "Hand Tool Cleaning" or SP 3 "Power Tool Cleaning"					

Notes for use:

- These products are designed primarily for overcoating and encapsulating existing coatings.
- A test patch, and subsequent adhesion measurements and observation of compatibility are recommended prior to specifying this type of coating. This reduces the risk of overcoating.
- Manufacturer's instructions are used here because there are a wide range of possibilities that require specific details of cleaning, coating and curing to address each individual circumstance.
- Areas of old coating that are feathered back should be built up to the topcoat level, before overcoating.
- If field steel has been exposed to weather, deicing salts or dirt, an SSPC- SP 12/NACE No. 5, LP WC water wash shall be done before blasting to remove salts to an SSPC- SP 12/NACE No. 5, NV-2 level.

STEEL, FIELD OVERCOAT - SYSTEM SFO3: Solvent-based System Single Component

MoT Usage: This system is designed for use in a field over coating situation where the original alkyd system is intact or where loose rust/paint can be removed by hand/power tools before repair and topcoating. It may be used in moderate to severe exposures.

PRIMER	SEALCOAT	STRIPECOAT	MIDCOAT	2nd MIDCOAT	TOPCOAT
As per Manufacturer's instructions	As per Manufacturer's instructions	As per Manufacturer's instructions	As per Manufacturer's instructions	None	As per manufacturer's instructions
Rusted areas and delaminated areas: High pressure water wash then SSPC- SP 11 "Powertool Cleaning to Bare Metal". Undamaged areas high pressure water wash					

Notes for use:

- The coatings listed under SF03 are too different to list a common application thickness.
- This system is designed as a two coat system.
- A test patch, and subsequent adhesion measurements and observation of compatibility are recommended prior to specifying this type of coating. This reduces the risk of overcoating.
- If field steel has been exposed to weather, deicing salts or dirt, an SSPC- SP 12/NACE No. 5, LP WC water wash shall be done before blasting to remove salts to an SSPC- SP 12/NACE No. 5, NV-2 level.

STEEL, FIELD OVERCOAT - SYSTEM SFO4: Solvent-based Surface Tolerant Epoxy

MoT Usage: This system is designed for use in a field over coating situation where the original alkyd system is intact or where loose rust/paint can be removed by hand/power tools. It may also be used in damp or condensing areas on structures. It may be used in moderate to severe exposures.

PRIMER	SEALCOAT	STRIPECOAT	MIDCOAT	2nd MIDCOAT	TOPCOAT
As per Manufacturer's instructions	None	As per Manufacturer's instructions	None	None	As per Manufacturer's instructions
Rusted areas and delaminated areas: Water wash as below then SSPC- SP 11 "Powertool Cleaning to Bare Metal". Undamaged areas water wash as below.					

Notes for use:

- This system is designed as a two coat system.
- This material shall not be applied to damp or condensing surfaces.
- Edges shall be feathered and where possible, bare areas shall be built up to the level of the previous topcoat before overcoating.
- If field steel has been exposed to weather, deicing salts or dirt, an SSPC- SP 12/NACE No. 5, LP WC water wash shall be done before blasting to remove salts to an SSPC- SP 12/NACE No. 5, NV-2 level.

STEEL, FIELD OVERCOAT - SYSTEM SFO5: Alkyd System on Steel Structures

MoT Usage: This system is designed for use in a field over coating situation where the original alkyd system is intact or where loose rust/paint can be removed by hand/power tools. It may be used for moderate exposures.

PRIMER	SEALCOAT	STRIPECOAT	MIDCOAT	2nd MIDCOAT	TOPCOAT
Alkyd Primer CAN/CGSB 1.40-89	None	Midcoat over prime	Alkyd Midcoat CAN/CGSB 1.59-89M	None	Alkyd Topcoat CAN/CGSB 1.59-89M
SSPC-SP 2 "Hand Tool Cleaning" or SSPC-SP 3 "Power Tool Cleaning"					

Notes for use:

- While this system will work on the cleaning regimen listed above it will perform better on SSPC-SP 6/ NACE No. 3 "Commercial blast".
- Edges shall be feathered and where possible, bare areas shall be built up to the level of the previous topcoat before overcoating.
- If field steel has been exposed to weather, deicing salts or dirt, an SSPC-SP 12/NACE No. 5, LP WC water wash shall be done before blasting to remove salts to an SSPC-SP 12/NACE No. 5, NV-2 level.

STEEL, PAINT CAULKING MATERIALS - SYSTEM PCM:

MoT Usage: These materials are designed to be used immediately under the topcoat material. They are compatible with most paint systems, but a test is recommended to ensure compatibility. They are used primarily to seal joints in steel structures to prevent the ingress of water in laps, splices etc. Manufacturer's instructions are followed for use.

Notes for use:

- These materials are applied on the midcoat, then coated with the final topcoat.
- Typically the top and sides of a plate are sealed and the bottom is not caulked to enable water to exit the crack if any gets in.
- If field steel has been exposed to weather, deicing salts or dirt, an SSPC- SP 12/NACE No. 5, LP WC water wash shall be done before blasting to remove salts to an SSPC- SP 12/NACE No. 5, NV-2 level.

STEEL, FIELD TOUCH-UP COATING SYSTEM - SYSTEM SFTA: Aerosol Products

MoT Usage: This system is designed for quick touch-up use in a field overcoating situation where the original alkyd system is intact or where loose rust/paint can be removed by hand/power tools. It is designed as a stopgap measure until proper maintenance coating can be scheduled.

PRIMER	SEALCOAT	STRIPECOAT	MIDCOAT	2nd MIDCOAT	TOPCOAT
Zinc Rich or Inhibitive alkyd	None	None	None	None	As needed
SSPC- SP 2 "Hand Tool Cleaning" or SP 3 "Power Tool Cleaning"					

Notes for use:

- All coatings will work better on a properly prepared surface (i.e., high pressure water washed and sandblasted).
- These products shall not be used on faying surfaces.

STEEL, FIELD TOUCH-UP COATING, BRUSHABLE PRODUCTS - SYSTEM SFTB:
Brushable Products

MoT Usage: This system is designed for brush application in a field touchup situation where the original alkyd system is intact and where loose rust/paint can be removed by hand/power tools.

PRIMER	SEALCOAT	STRIPECOAT	MIDCOAT	2nd MIDCOAT	TOPCOAT
Alkyd Primer CAN/CGSB 1.40-89 Zinc Rich alkyd	None	Midcoat over prime	Alkyd Midcoat CAN/CGSB 1.59-89M	None	Alkyd Topcoat CAN/CGSB 1.59-89M
SSPC- SP 2 "Hand Tool Cleaning" or SP 3 "Power Tool Cleaning" or SP 11 Industrial					

Notes for use:

- These products require different surface preparation and application thickness, refer to manufacturer's literature.
- All coatings will work better on a properly prepared surface (i.e., high pressure water washed and sandblasted).
- These products shall not be used on faying surfaces.
- If field steel has been exposed to weather, deicing salts or dirt, an SSPC- SP 12/NACE No. 5, LP WC water wash shall be done before blasting to remove salts to an SSPC- SP 12/NACE No. 5, NV-2 level.

STEEL, PENETRATING SEALERS - SYSTEM SPS.

MoT Usage: These products are designed for use in the field to help seal joints, pack rusted areas, riveted and bolted areas, and generally any area where a product that will wick into any small cracks and crevices to help seal them. These products must be topcoated as most are epoxy based and will not provide good service in exterior sunlight.

PRIMER	SEALCOAT	STRIPECOAT	MIDCOAT	2nd MIDCOAT	TOPCOAT
If Used As per Manufacturer's Instructions	As per Manufacturer's Instructions	If Used As per Manufacturer's Instructions	If Used As per Manufacturer's Instructions		As per Manufacturer's Instructions

Notes for use:

- It is important to apply these products exactly as the Manufacturer states since they do not tolerate over thickness well.
- These products shall be used only with compatible products from the same manufacturer
- When used as part of a coating system, these products shall be applied after the primer coat, before the stripe and midcoats. This is to take greatest advantage of the sacrificial primer which must be in electrical contact with the steel to be most effective.

PRODUCTS FOR UNIQUE SITUATIONS: ENCLOSED SPACES AND PROTECTED AREAS - SYSTEM U1: Products for Enclosed Spaces and Protected Areas

MoT Usage: This system is designed to use in situations where corrosion protection is desired and where high gloss is not as important. It is not for use in abrasive areas or where dirt retention is a problem.

PRIMER	SEALCOAT	STRIPECOAT	MIDCOAT	2nd MIDCOAT	TOPCOAT
	Penetrating Sealer	Topcoat material if needed	Topcoat material if needed	None	As Per Manufacturer's Instructions

Notes for use:

- The penetrating sealer shall be used to penetrate and seal rusted areas and lap joints that cannot be properly cleaned. This is a unique application and Manufacturer's instructions shall be followed.
- The topcoat is a relatively soft, inhibitive coating that is surface tolerant and application tolerant. It does not harden enough to place in areas that are abrasive, traveled, or accessible to the public.
- The sealer may be used before the primer or any of the successive coats. Typically the sealer is used and then coated with a mid or topcoat

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PRODUCTS FOR UNIQUE SITUATIONS: LOW TEMPERATURE / HIGH HUMIDITY - SYSTEM U2: Products for low temperature / high humidity use

MoT Usage: This system is designed to use in situations where low temperature and/or high humidity preclude the use of other coating systems**.**

PRIMER	SEALCOAT	STRIPECOAT	MIDCOAT	2nd MIDCOAT	TOPCOAT
As Per Manufacturer's Instructions	As Per Manufacturer's Instructions	As Per Manufacturer's Instructions	As Per Manufacturer's Instructions	None	As Per Manufacturer's Instructions

Notes for use:

- These coatings are moisture cured, therefore the entire can shall be used at once to prevent reaction of the coating in the can resulting from contact with atmospheric moisture.
- These coatings may also be used at normal coating temperatures and humidities. Since it is cured by reaction with atmospheric moisture, it should not be used in low humidity cure situations.
- If field steel has been exposed to weather, deicing salts or dirt, an SSPC-SP 12/NACE No. 5, LP WC water wash shall be done before blasting to remove salts to an SSPC-SP 12/NACE No. 5, NV-2 level.

STEEL, RAILINGS, HOT DIP GALVANIZING - SYSTEM SRG: Hot Dip Galvanizing Coating of Removable Rails

MoT Usage: This system is designed for use where small parts such as railing can be removed and sent to the galvanizing plant.

PRIMER	SEALCOAT	STRIPECOAT	MIDCOAT	2nd MIDCOAT	TOPCOAT
None	None	None	None	None	Zinc
Surface preparation is done at the galvanizing plant.					

Notes for use:

- This is the preferred treatment for rails as it provides a very long life combined with low maintenance.
- The colour is grey. A colour coat is not recommended due to the increased capital costs and maintenance.
- Sections with closed tubes shall not be done without relieving the tubes by drilling holes to let the hot gases escape. This shall be designed into the item.
- Both the inside and outside of sections are coated with zinc.
- Defects include inclusions, dross, runs and mottling.

STEEL, RAILINGS AND POSTS, METALLIZING - SYSTEM SRM: Metallizing For Steel Railings And Posts

MoT Usage: This system is used primarily to coat steel fence posts that cannot be easily removed from a structure. It can however be used to coat any structural element in place. For elements which can be removed, hot dip galvanizing is the preferred treatment.

PRIMER	SEALCOAT	STRIPECOAT	MIDCOAT	2nd MIDCOAT	TOPCOAT
None	None	None	None	None	Zinc
SSPC-SP 5 /NACE No. 1 White metal, Sharp Profile: 50 - 75 µm (2 - 3 mils)					

Notes for use:

- Substrate shall be very clean.
- Zinc or zinc:aluminum (85:15) wire or powder shall be used.
- If field steel has been exposed to weather, deicing salts or dirt, an SSPC-SP 12/NACE No. 5, LP WC water wash shall be done before blasting to remove salts to an SSPC-SP 12/NACE No. 5, NV-2 level.
- This coating may be overcoated to increase lifetime, although it is not necessary